

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Eighth Semester of B. Tech. (CL) Examination

November 2013

CL 431 Earthquake Engineering

Date: 29.11.2013, Friday

Time: 10:00 a.m. To 01:00 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures with pencil only wherever required.
4. Use of scientific calculator is allowed.
5. IS 1893: 2002 Part-I, IS 4326: 1993 & IS 13920: 1993 are permitted.

SECTION – I

- Q – 1 (a) Draw and detail the typical qualitative reinforcement detailing of two span reinforced concrete continuous rectangular beam of dimension 230 mm × 500 mm as per IS 13920:1993. [03]
- (b) What is ductile detailing of members? Explain the same for RC beams as per provision of IS 13920-1993. [04]
- Q – 2 (a) Specify the requirements of special confining reinforcement for column, as per IS 13920:1993. [02]
- (b) Attempt any Two: [12]
1. A vibrating system consists of a mass of 50 kg, a spring of stiffness 30 kN/m and a viscous damper. For the following conditions, find the equation for displacement at any instant of time. Assume critically damped system. The initial displacement and velocity given as 5 cm and 0 m/sec respectively, and the system is released. Also find the time taken by the mass to reach at 5 mm from equilibrium position on return after releasing the system.
 2. An empty elevated water tank is pulled by a steel cable by applying a 30 kN force horizontally and displaced by 5 cm. the cable is suddenly cut and the resulting free vibration is recorded. At the end of five complete cycles, the time is 2.0 second and the amplitude is 2 cm. determine the damping ratio, natural period of undamped vibration, effective stiffness, weight and damping coefficient.
 3. A water tank with container filled with water weigh 450 kN and resting on a shaft having stiffness 4.38×10^7 N/m. it is subjected to ground motion produced by passing train in the vicinity of tank. The ground motion is idealized as harmonic acceleration of foundation of tower with amplitude of 0.1 g at a frequency of 10 cycles/sec. determine position of tower related to the motion of its foundation. If effective

damping coefficient is 10% of critical damping. Also determine the transmissibility of the displacement.

Q – 3 (a) Sketch the reinforcement detailing for cross section of RCC column 400 mm × 400 mm, having 8 nos 20 mm diameter main bars as ductile requirement. [02]

(b) Attempt any One: [12]

1. For three storey building frame, lumped mass and storey stiffness are shown in **Figure 1**. Find out lateral forces at each floor using CQC method. Take mass in kg and stiffness of column, $k = 5 \times 10^5$ N/m. Assume 5% damping and hard soil is available at foundation level. Consider Zone – V and the given building is a school building. Take $R=5$.
2. For three storey building frame, lumped mass and storey stiffness are shown in **Figure 1**. Find out lateral forces at each floor using SRSS method. Take mass in kg and stiffness of column, $k = 5 \times 10^5$ N/m. Assume 5% damping and hard soil is available at foundation level. Consider Zone – V and the given building is a school building. Take $R=5$.

SECTION – II

Q – 4 (a) Enlist the major tectonic plates of India. [02]

(b) What is fault? Explain different types of faults with neat sketches. [05]

Q - 5 Attempt any Two: [14]

- (a) Calculate the torsional shear forces in one-storey shear wall masonry structure with a rigid diaphragm roof as shown in **Figure 2**. The relative rigidity of each shear wall is given. All walls are a total of 5 m height, 4 m up to roof level and 1 m parapet Seismic Zone V, $Z=0.36$, $I= 1.0$, $Sa/g = 2.5$, $R= 1.5$
- (b) Determine the lateral forces on one-storey unreinforced brick masonry building situated at Allahabad (zone III).

Building Data:

Plan size = 18 m × 8 m

Height of building = 3.1 m

Weight of roof = 2.5 kN/m²

Weight of walls = 3.0 kN/m²

Live load at roof = 0

Live load at floors = 1 kN/m²

Zone factor (z) = 0.10

Importance factor (I) = 1.0

Response reduction factor (R) = 1.5

Soil: (Type-II) Medium soil

- (c) Define bands. At what levels in a masonry building would you provide them? Give justifications for each of them.

For a building shown in Figure 3, locate the center of mass.

Q - 6 Attempt any Two:

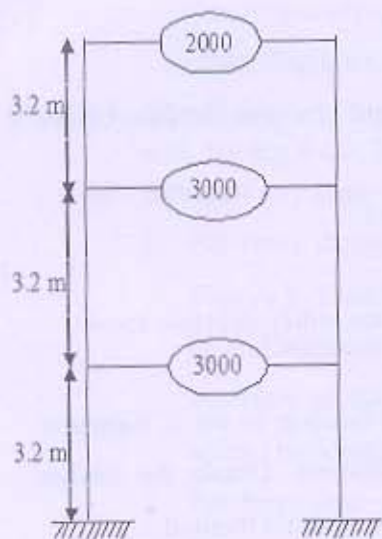
[14]

- (a) Determine the natural frequencies and mode shapes of the three-storey structure shown in Figure 4.
- (b) For building of four-storey, the dynamic properties for vibration in the X direction have been obtained by carrying out a free vibration analysis. Obtain the design seismic force for Mode 1 in the X direction by the dynamic analysis method.

Free vibration properties of the building for vibration in the X direction

Storey's	Time period (sec)	Mode 1
		0.860
	Weight W (kN)	Mode Shape
Roof	3000	1.000
3 rd Floor	4200	0.904
2 nd Floor	4200	0.716
1 st Floor	4200	0.441

- (c) Consider a four-storey reinforced concrete office building shown in Figure 5. The building is located in Shillong (seismic zone V). The soil conditions are medium stiff and the entire building is supported on a raft foundation. The R. C. frames are infilled with brick-masonry. The lumped weight due to dead loads is 12 kN/m^2 on floors and 10 kN/m^2 on the roof. The floors are to cater for a live load of 4 kN/m^2 on floors and 1.5 kN/m^2 on the roof. Determine design seismic load on the structure.



$T_1 = 0.992 \text{ sec}$
 $T_2 = 0.362 \text{ sec}$
 $T_3 = 0.262 \text{ sec}$
 Mode shapes are:
 $\phi_1 = [1.0; 1.759; 2.095]$
 $\phi_2 = [1.0; 0.196; -0.961]$
 $\phi_3 = [1.0; -1.456; 1.12]$

Figure 1

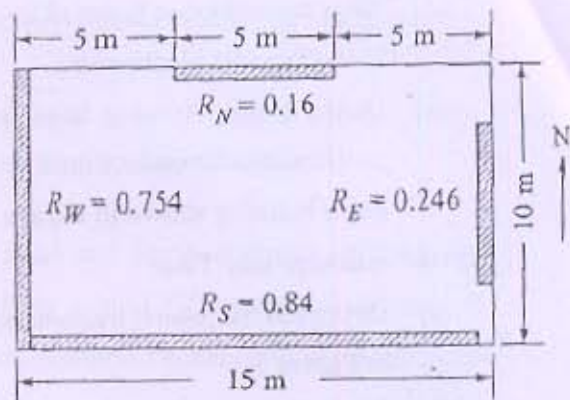


Figure 2

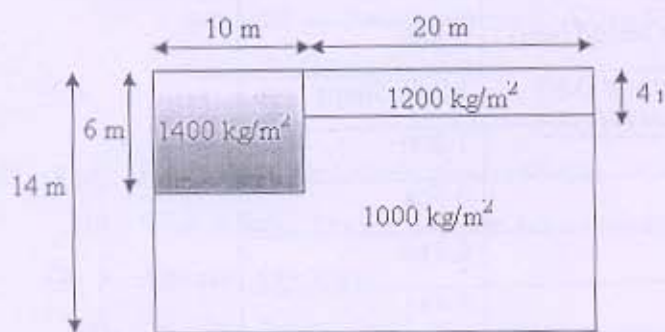


Figure 3

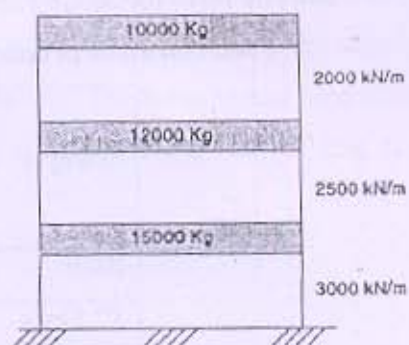


Figure 4

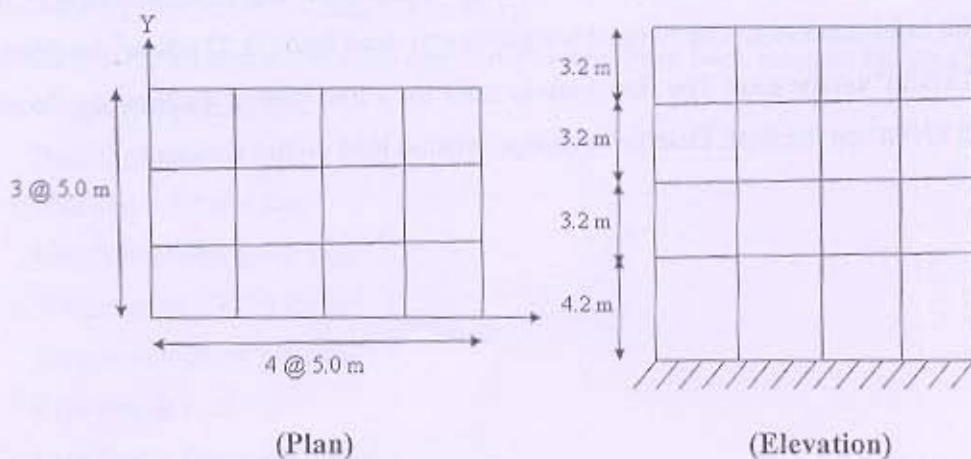


Figure 5
